Lecture 5. Ensemble Learning

Crowd intelligence

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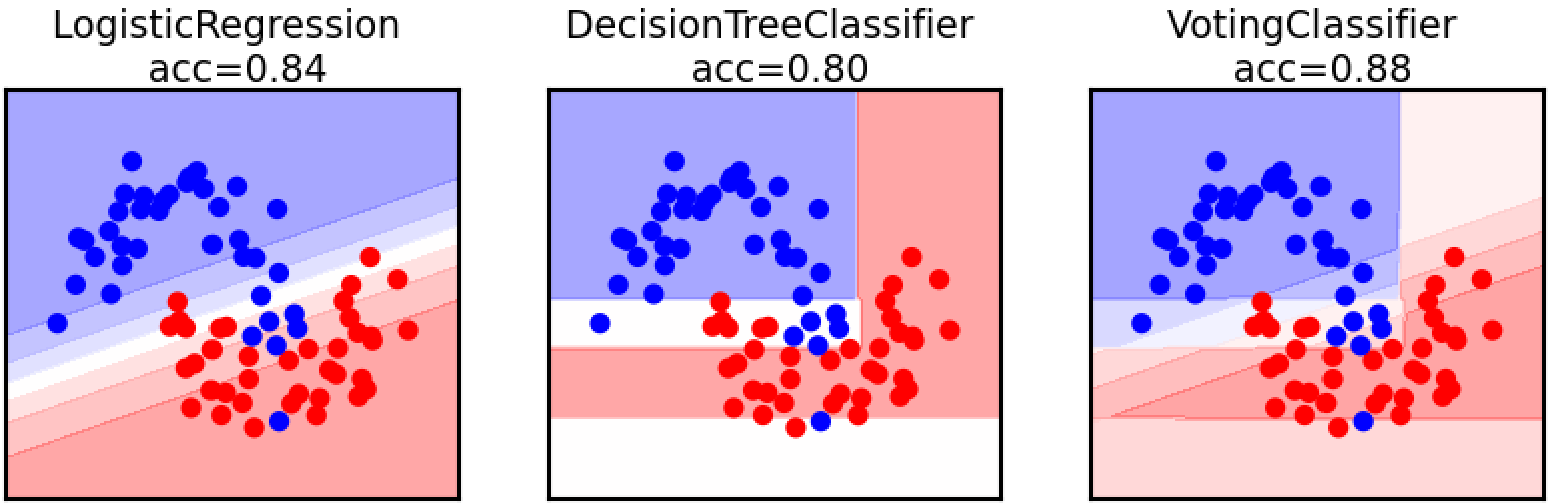
Ensemble learning

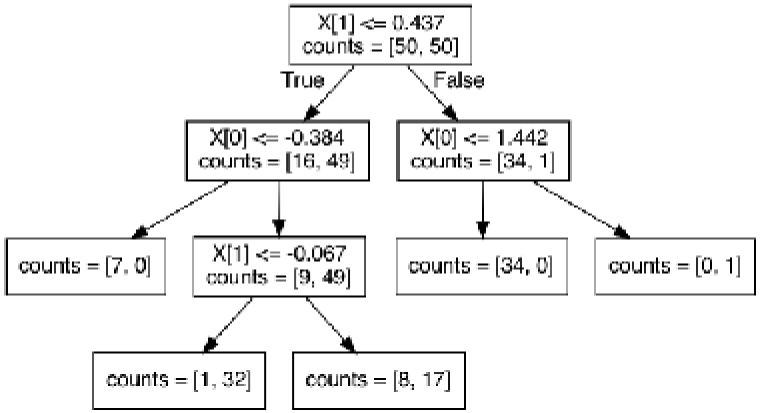
- If different models make different mistakes, can we simply average the predictions?
- Voting Classifier: gives every model a *vote* on the class label
    - Hard vote: majority class wins (class order breaks ties)
    - Soft vote: sum class probabilities $p_{m,c}$ over M models: $\underset{c}{\operatorname{argmax}} \sum_{m=1}^{M} w_c p_{m,c}$
    - Classes can get different weights w_c (default: $w_c = 1$)

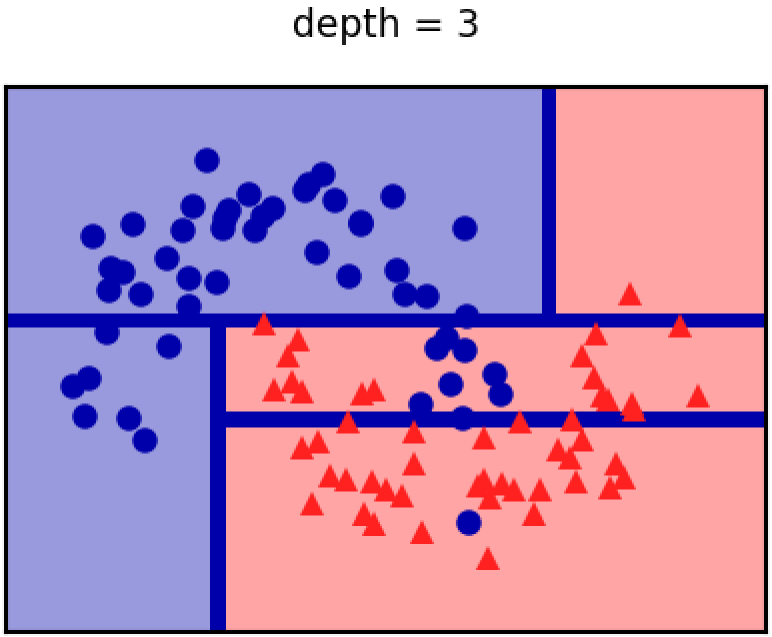


- Why does this work?
    - Different models may be good at different 'parts' of data (even if they underfit)
    - Individual mistakes can be 'averaged out' (especially if models overfit)
- Which models should be combined?
- Bias-variance analysis teaches us that we have two options:
    - If model underfits (high bias, low variance): combine with other low-variance models
        - Need to be different: 'experts' on different parts of the data
        - Bias reduction. Can be done with ***Boosting***
    - If model overfits (low bias, high variance): combine with other low-bias models
        - Need to be different: individual mistakes must be different
        - Variance reduction. Can be done with ***Bagging***
- Models must be uncorrelated but good enough (otherwise the ensemble is worse)
- We can also *learn* how to combine the predictions of different models: ***Stacking***

Decision trees (recap)

- Representation: Tree that splits data points into leaves based on tests
- Evaluation (loss): Heuristic for purity of leaves (Gini index, entropy,...)
- Optimization: Recursive, heuristic greedy search (Hunt's algorithm)
    - Consider all splits (thresholds) between adjacent data points, for every feature
    - Choose the one that yields the purest leafs, repeat

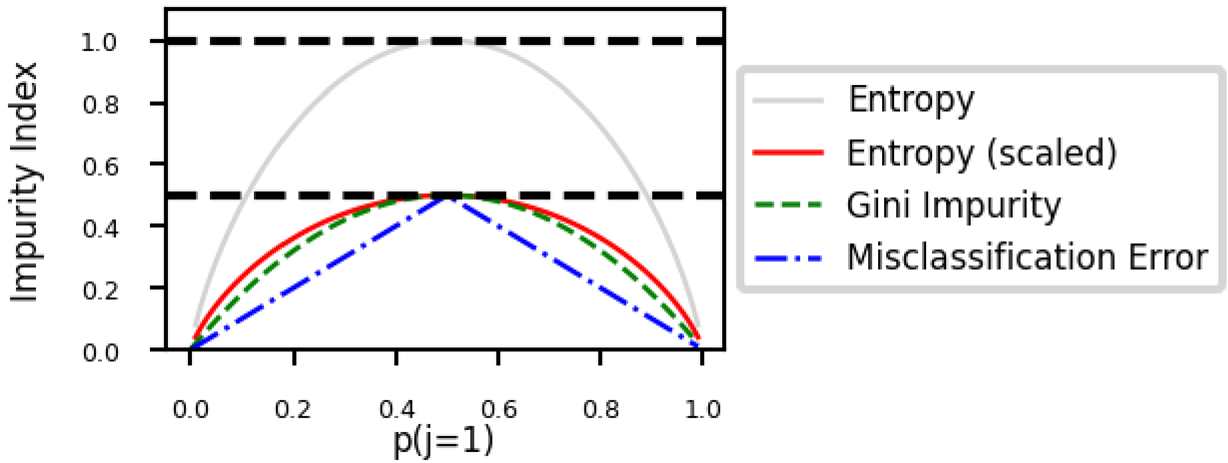




Evaluation (loss function for classification)

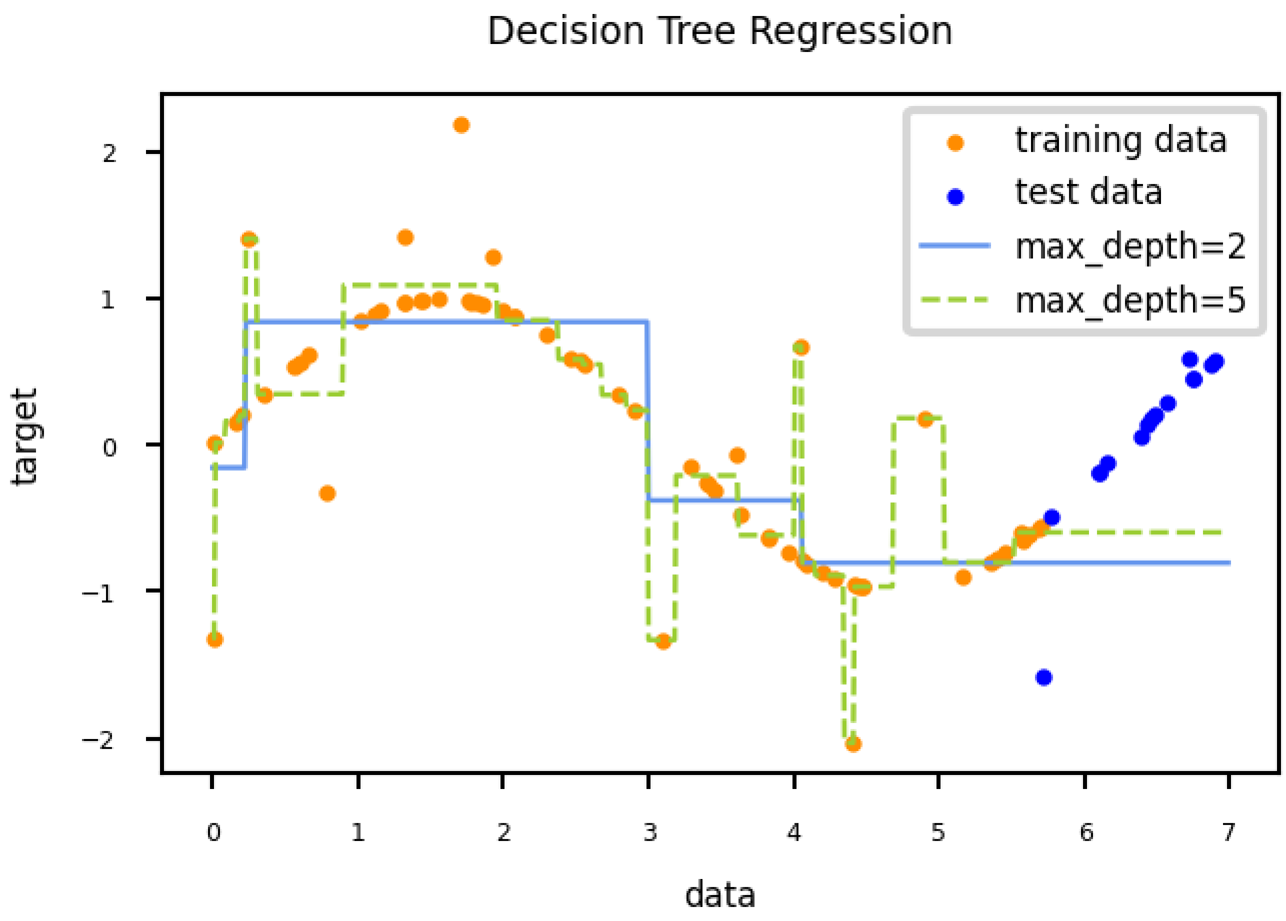
- Every leaf predicts a class probability $\hat{p}_c$ = the relative frequency of class c
- Leaf impurity measures (splitting criteria) for L leafs, leaf l has data X_l:
  - Gini-Index: $Gini(X_l) = \sum_{c \neq c'} \hat{p}_c \hat{p}_{c'}$ Gini=preferred because easier to compute
  - Entropy (more expensive): $E(X_l) = -\sum_{c \neq c'} \hat{p}_c \log_2 \hat{p}_c$
  - Best split maximizes *information gain* (idem for Gini index)

$$Gain(X, X_i) = E(X) - \sum_{l=1}^{L} \frac{|X_{i=l}|}{|X_i|} E(X_{i=l})$$



Regression trees

- Every leaf predicts the *mean* target value μ of all points in that leaf
- Choose the split that minimizes squared error of the leaves: $\sum_{x_i \in L} (y_i - \mu)^2$
- Yields non-smooth step-wise predictions, cannot extrapolate

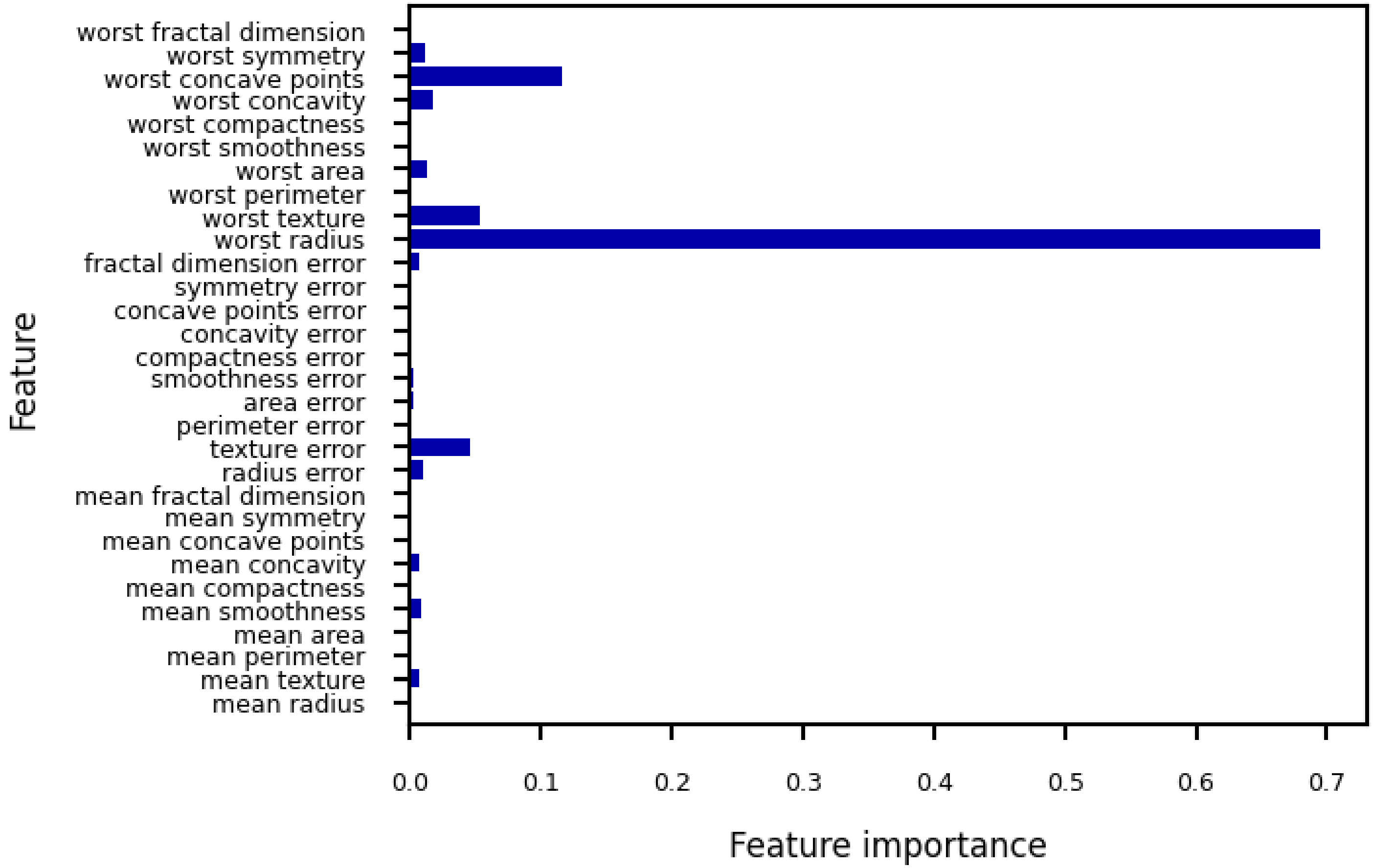


If test data is not in same distribution as train data, then bad predictions (cannot extrapolate). So never use when test data is outside range of train data

Often used in boosting algorithms.

Impurity/Entropy-based feature importance

- We can measure the importance of features (to the model) based on
    - Which features we split on
    - How high up in the tree we split on them (first splits ar emore important)



If splitting a feature makes prediction easier, then feature tells us something about target we want to predict.

Thus good way to estimate how useful a feature is.

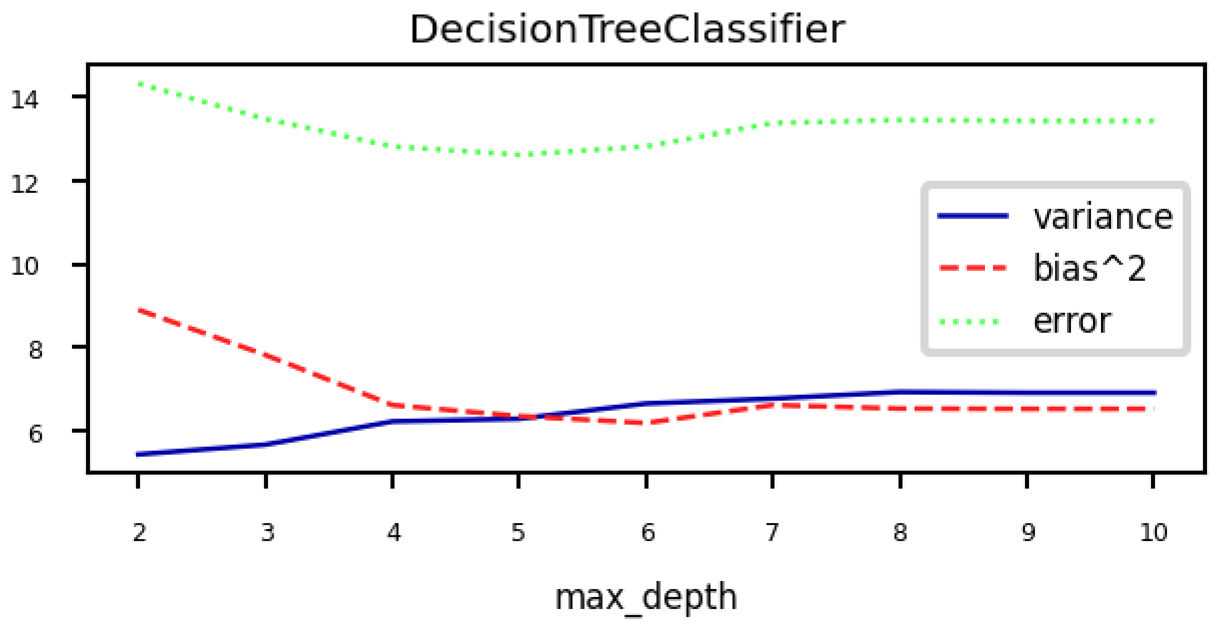
Every split happens on a feature

All features that are being split on are important (and higher up = more important).

However, vulnerable to small changes in training points. Many trees and average over that = good indication of feature importance.

Under- and overfitting

- We can easily control the (maximum) depth of the trees as a hyperparameter
- Bias-variance analysis:
  - Shallow trees have high bias but very low variance (underfitting)
  - Deep trees have high variance but low bias (overfitting)
- Because we can easily control their complexity, they are ideal for ensembling
  - Deep trees: keep low bias, reduce variance with ***Bagging***
  - Shallow trees: keep low variance, reduce bias with ***Boosting***



Bagging (Bootstrap Aggregating)

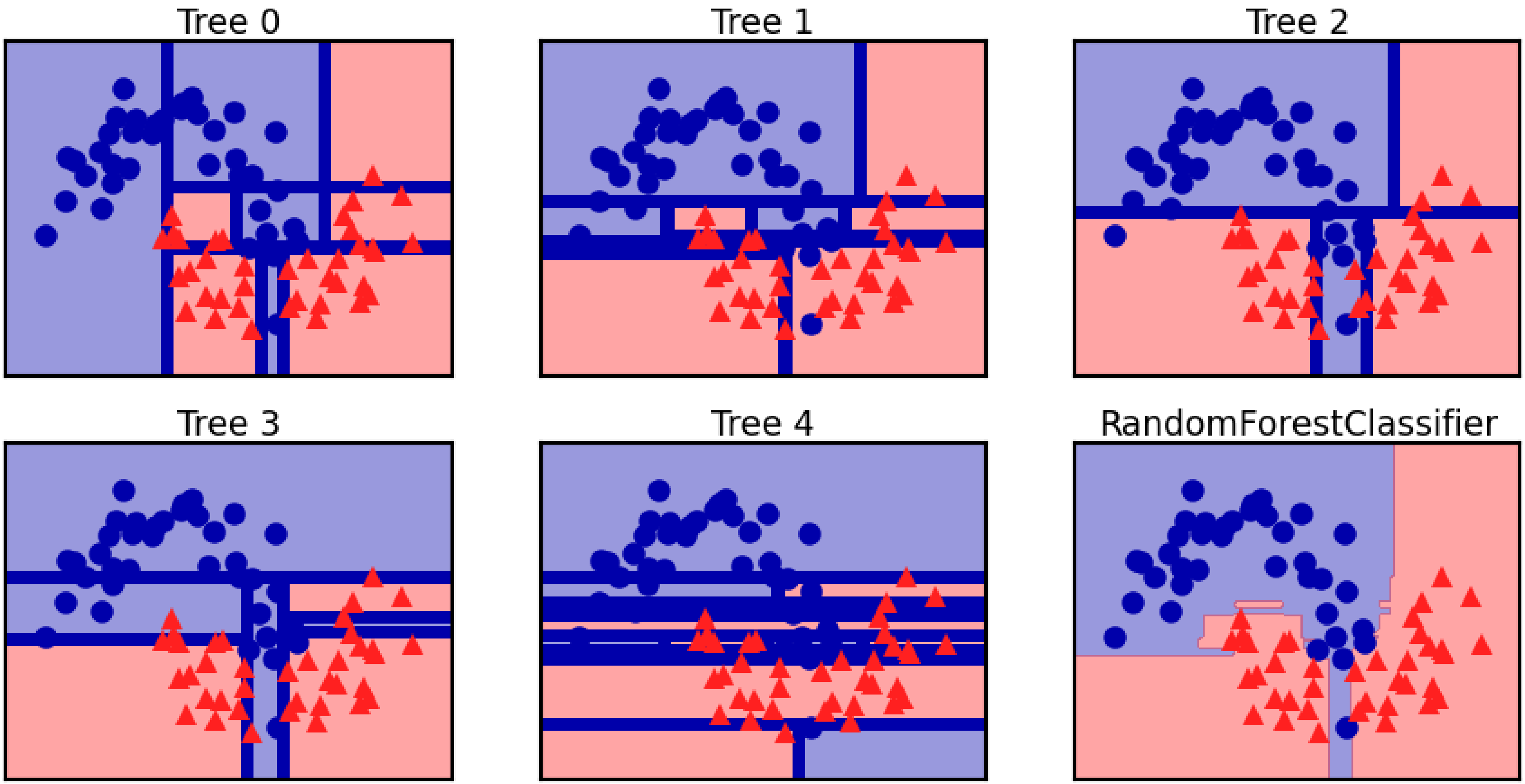
- Obtain different models by training the *same* model on *different training samples*
    - Reduce overfitting by averaging out individual predictions (variance reduction)
- In practice: take I bootstrap samples of your data, train a model on each bootstrap
    - Higher I: more models, more smoothing (but slower training and prediction)
- Base models should be unstable: different training samples yield different models
    - E.g. very deep decision trees, or even randomized decision trees
    - Deep Neural Networks can also benefit from bagging (deep ensembles)
- Prediction by averaging predictions of base models
    - Soft voting for classification (possibly weighted)
    - Mean value for regression
- Can produce uncertainty estimates as well
    - By combining class probabilities of individual models (or variances for regression)

Random Forests

Extreme random trees = smoother predictions

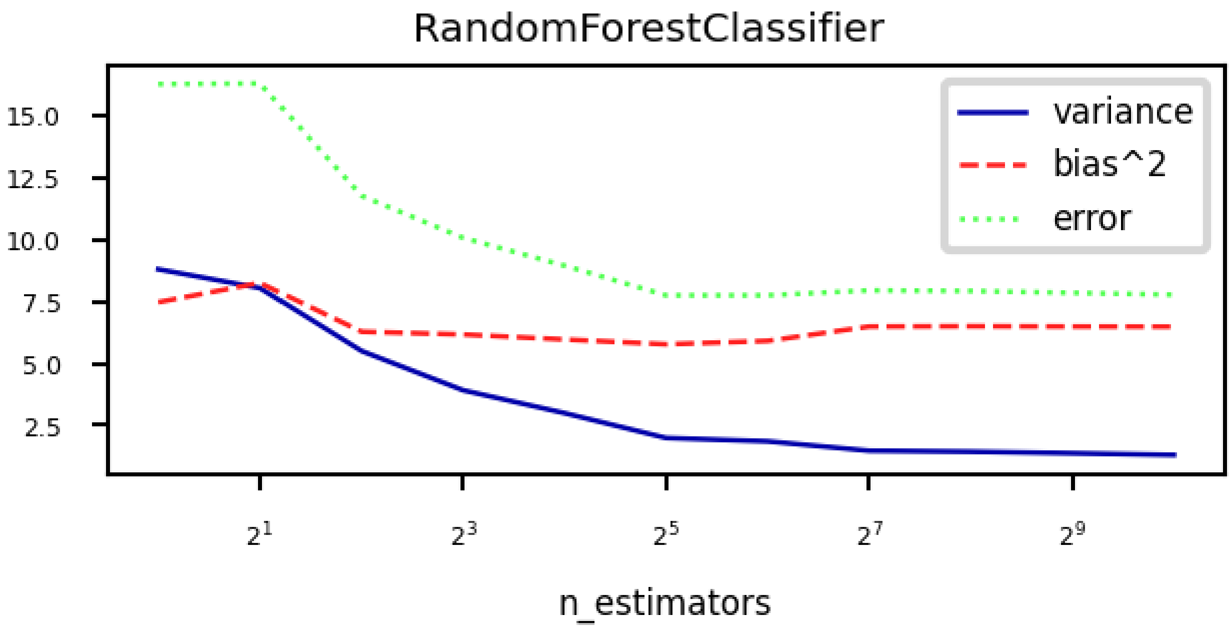
- Uses *randomized trees* to make models even less correlated (more unstable) High variance, low bias
  - At every split, only consider `max_features` features, randomly selected
- Extremely randomized trees: considers 1 random threshold for random set of features (faster)

All very different overfitted trees Then average out over all predictions Extreme: take one random split point for each feature, and then take the best



Effect on bias and variance

- Increasing the number of models (trees) decreases variance (less overfitting)
- Bias is mostly unaffected, but will increase if the forest becomes too large (oversmoothing)

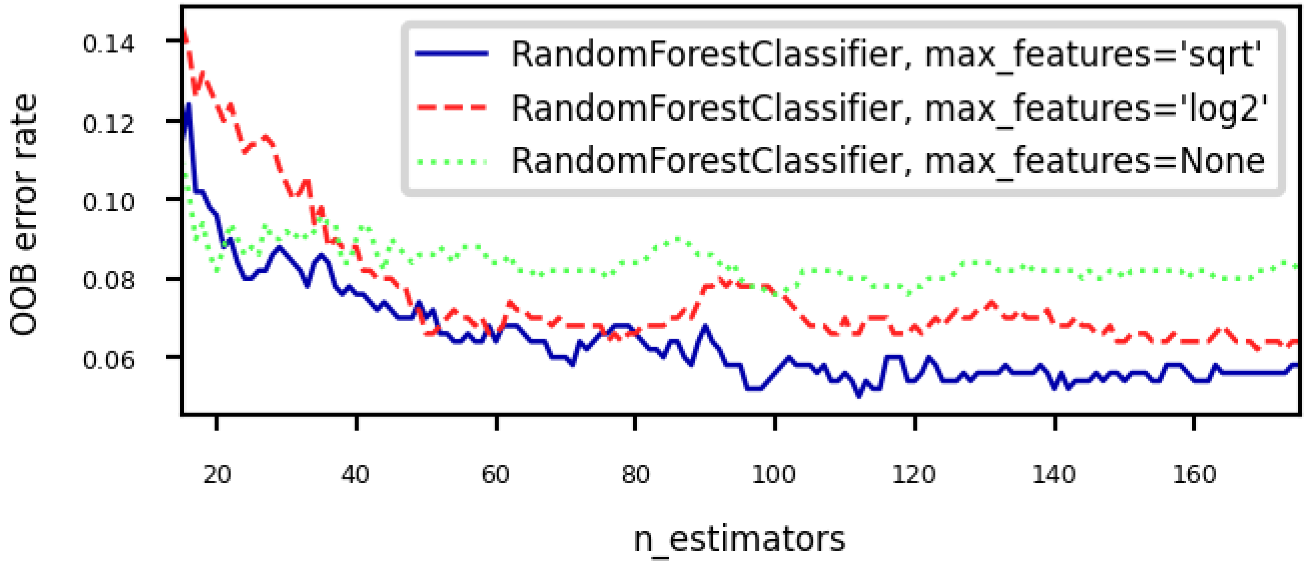


In practice

- Different implementations can be used. E.g. in scikit-learn:
    - `BaggingClassifier`: Choose your own base model and sampling procedure
    - `RandomForestClassifier`: Default implementation, many options
    - `ExtraTreesClassifier`: Uses extremely randomized trees
- Most important parameters:
    - `n_estimators` (>100, higher is better, but diminishing returns)
        - Will start to underfit (bias error component increases slightly)
    - `max_features`
        - Defaults: $sqrt(p)$ for classification, $log2(p)$ for regression
        - Set smaller to reduce space/time requirements
    - parameters of trees, e.g. `max_depth`, `min_samples_split`,...
        - Prepruning useful to reduce model size, but don't overdo it
- Easy to parallelize (set `n_jobs` to -1)
- Fix `random_state` (bootstrap samples) for reproducibility

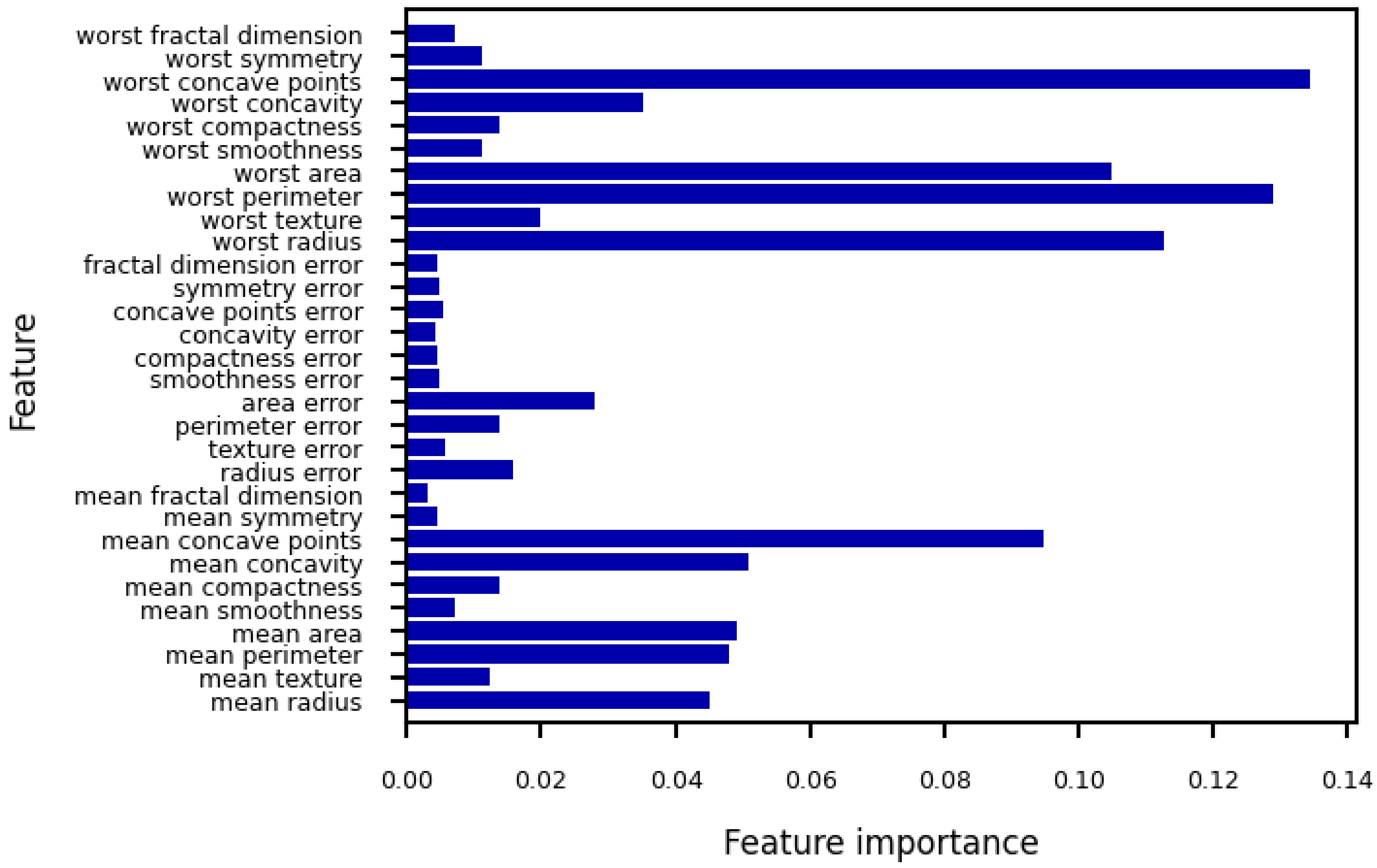
Out-of-bag error

- RandomForests don't need cross-validation: you can use the out-of-bag (OOB) error
- For each tree grown, about 33% of samples are out-of-bag (OOB)
    - Remember which are OOB samples for every model, do voting over these
- OOB error estimates are great to speed up model selection
    - As good as CV estimates, althought slightly pessimistic
- In scikit-learn: `oob_error = 1 - clf.oob_score_`



Feature importance

- RandomForests provide more reliable feature importances, based on many alternative hypotheses (trees)



Other tips

- Model calibration
    - RandomForests are poorly calibrated.
    - Calibrate afterwards (e.g. isotonic regression) if you aim to use probabilities
- Warm starting
    - Given an ensemble trained for I iterations, you can simply add more models later
    - You *warm start* from the existing model instead of re-starting from scratch
    - Can be useful to train models on new, closely related data
        - Not ideal if the data batches change over time (concept drift)
        - Boosting is more robust against this (see later)

Strength and weaknesses

- RandomForest are among most widely used algorithms:
    - Don't require a lot of tuning
    - Typically very accurate
    - Handles heterogeneous features well (trees)
    - Implictly selects most relevant features
- Downsides:
    - less interpretable, slower to train (but parallellizable)
    - don't work well on high dimensional sparse data (e.g. text)

Adaptive Boosting (AdaBoost)

- Obtain different models by *reweighting* the training data every iteration
    - Reduce underfitting by focusing on the 'hard' training examples
- Increase weights of instances misclassified by the ensemble, and vice versa
- Base models should be simple so that different instance weights lead to different models
    - Underfitting models: decision stumps (or very shallow trees)
    - Each is an 'expert' on some parts of the data
- Additive model: Predictions at iteration I are sum of base model predictions
    - In Adaboost, also the models each get a unique weight w_i

$$f_I(\mathbf{x}) = \sum_{i=1}^{I} w_i g_i(\mathbf{x})$$

- Adaboost minimizes exponential loss. For instance-weighted error ε:

$$\mathcal{L}_{Exp} = \sum_{n=1}^{N} e^{\varepsilon(f_I(\mathbf{x}))}$$

- By deriving $\frac{\partial \mathcal{L}}{\partial w_i}$ you can find that optimal $w_i = \frac{1}{2}\log(\frac{1-\varepsilon}{\varepsilon})$

AdaBoost algorithm

- Initialize sample weights: $s_{n,0} = \frac{1}{N}$
- Build a model (e.g. decision stumps) using these sample weights
- Give the *model* a weight w_i related to its weighted error rate ε

$$w_i = \lambda \log(\frac{1-\varepsilon}{\varepsilon})$$

    - Good trees get more weight than bad trees
    - Logit function maps error ε from [0,1] to weight in [-Inf,Inf] (use small minimum error)
    - Learning rate λ (shrinkage) decreases impact of individual classifiers
        - Small updates are often better but requires more iterations
- Update the sample weights
    - Increase weight of incorrectly predicted samples: $s_{n,i+1} = s_{n,i}e^{w_i}$
    - Decrease weight of correctly predicted samples: $s_{n,i+1} = s_{n,i}e^{-w_i}$
    - Normalize weights to add up to 1
- Repeat for I iterations

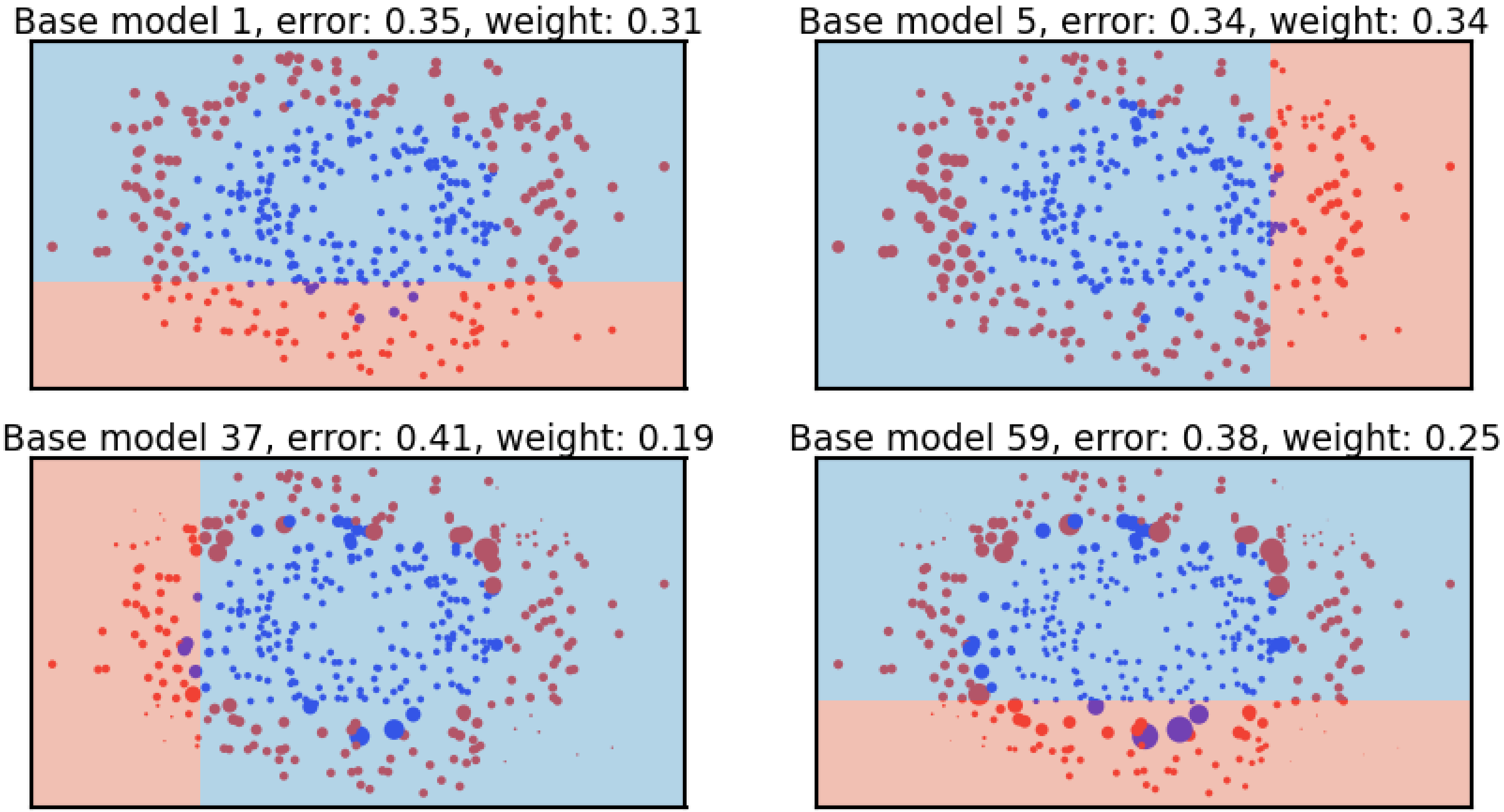
AdaBoost variants

- Discrete Adaboost: error rate ε is simply the error rate (1-Accuracy)
- Real Adaboost: ε is based on predicted class probabilities $\hat{p}_c$ (better)
- AdaBoost for regression: ε is either linear ($|y_i - \hat{y}_i|$), squared ($(y_i - \hat{y}_i)^2$), or exponential loss
- GentleBoost: adds a bound on model weights w_i
- LogitBoost: Minimizes logistic loss instead of exponential loss

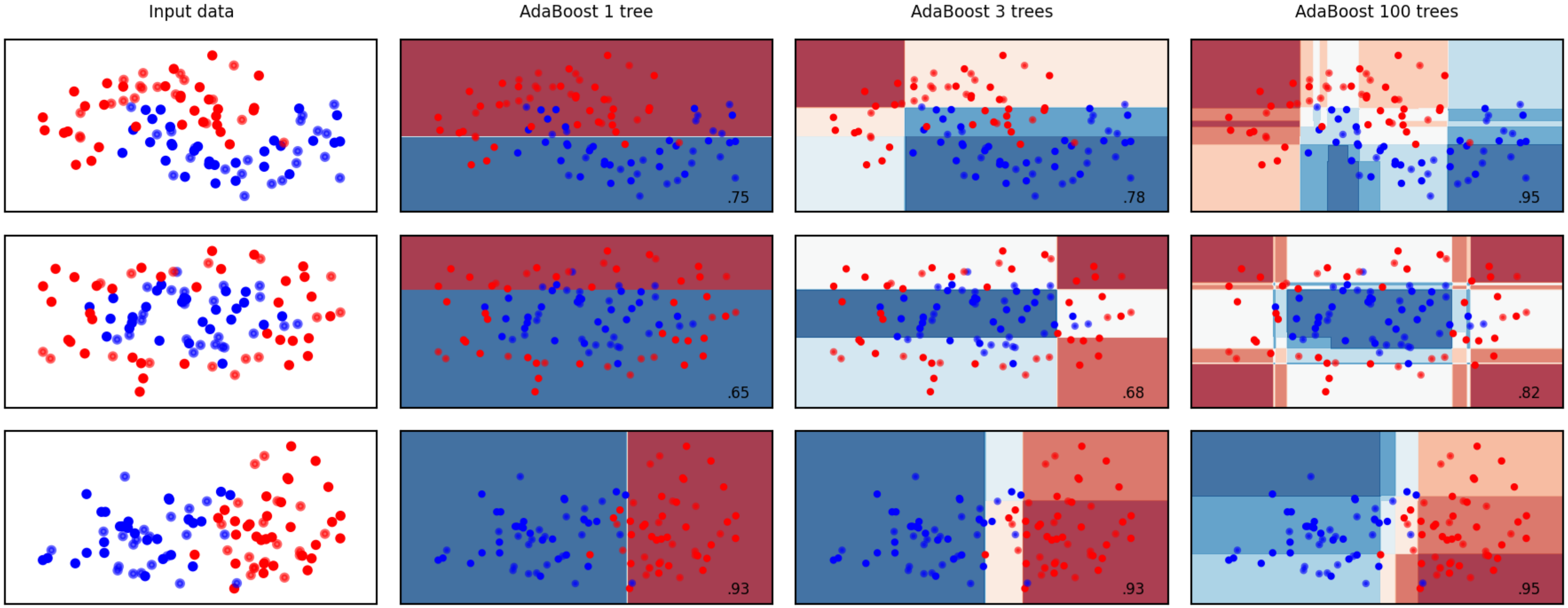
$$\mathcal{L}_{Logistic} = \sum_{n=1}^{N} log(1 + e^{\varepsilon(f_I(\mathbf{x}))})$$

Adaboost in action

- Size of the samples represents sample weight
- Background shows the latest tree's predictions



Examples


Input data
AdaBoost 1 tree
AdaBoost 3 trees
AdaBoost 100 trees
.75
.78
.95
.65
.68
.82
.93
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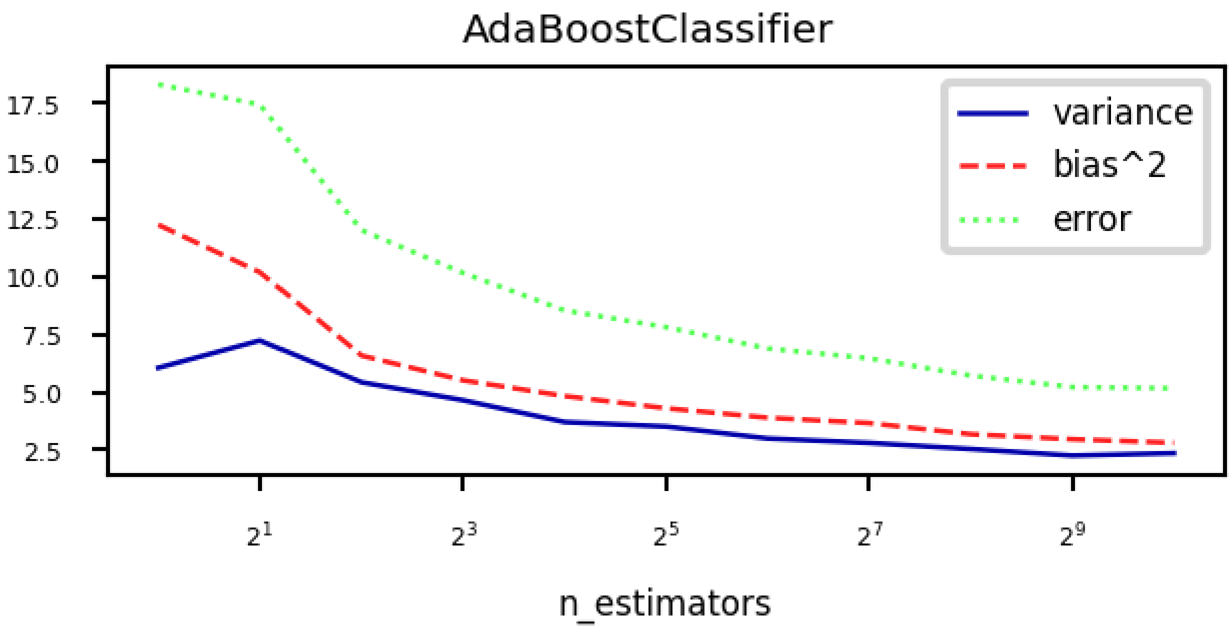
.95

Bias-Variance analysis

- AdaBoost reduces bias (and a little variance)
    - Boosting is a *bias reduction* technique
- Boosting too much will eventually increase variance



Gradient Boosting

- Ensemble of models, each fixing the remaining mistakes of the previous ones
    - Each iteration, the task is to predict the *residual error* of the ensemble
- Additive model: Predictions at iteration I are sum of base model predictions
    - Learning rate (or *shrinkage*) η: small updates work better (reduces variance)

$$f_I(\mathbf{x}) = g_0(\mathbf{x}) + \sum_{i=1}^{I} \eta \cdot g_i(\mathbf{x}) = f_{I-1}(\mathbf{x}) + \eta \cdot g_I(\mathbf{x})$$

- The *pseudo-residuals* r_i are computed according to differentiable loss function
    - E.g. least squares loss for regression and log loss for classification
    - Gradient descent: *predictions* get updated step by step until convergence

$$g_i(\mathbf{x}) \approx r_i = -\frac{\partial \mathcal{L}(y_i, f_{i-1}(x_i))}{\partial f_{i-1}(x_i)}$$

- Base models g_i should be low variance, but flexible enough to predict residuals accurately
    - E.g. decision trees of depth 2-5

Gradient Boosting Trees (Regression)

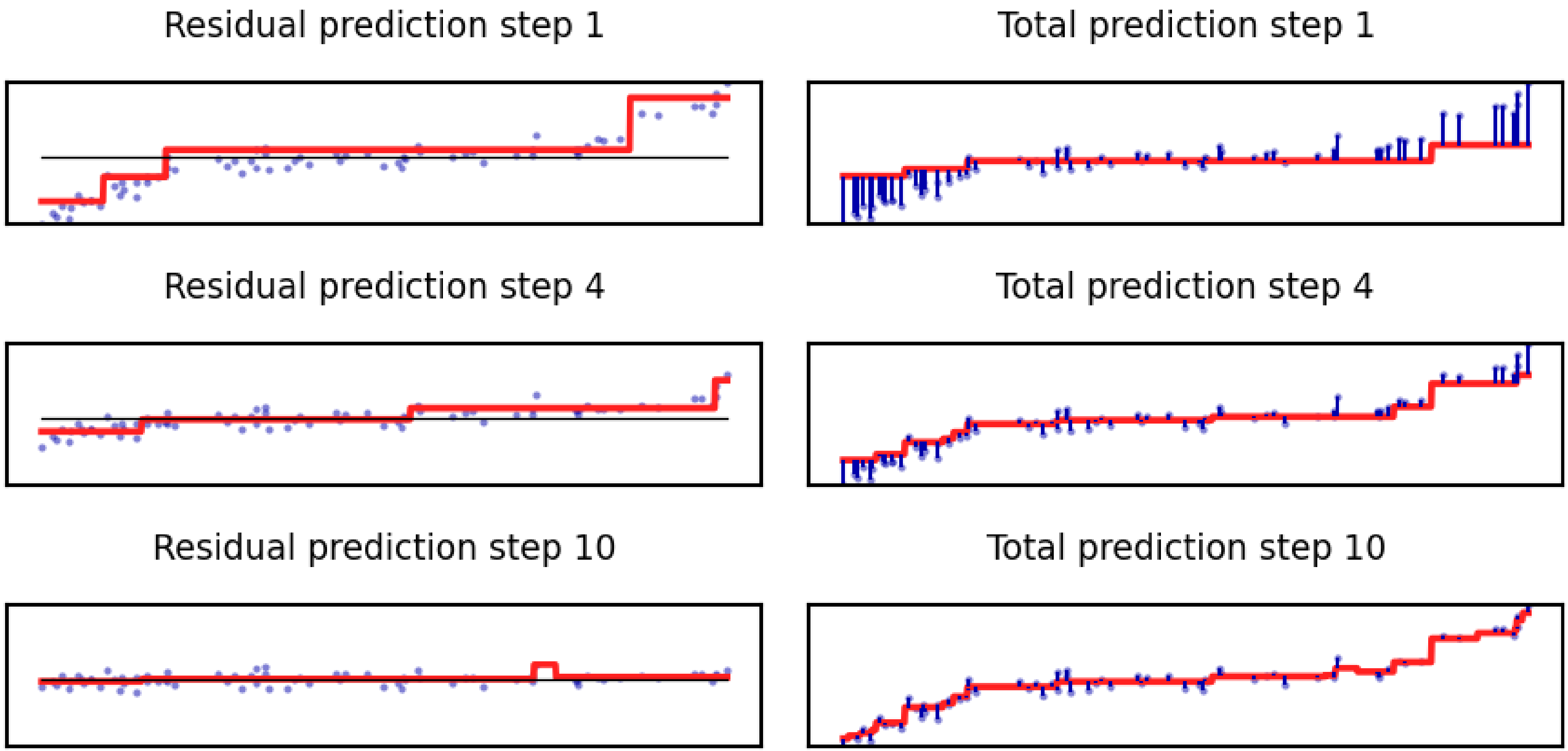
- Base models are regression trees, loss function is square loss: $\mathcal{L} = \frac{1}{2}(y_i - \hat{y}_i)^2$
- The pseudo-residuals are simply the prediction errors for every sample:

$$r_i = -\frac{\partial \mathcal{L}}{\partial \hat{y}} = -2 * \frac{1}{2}(y_i - \hat{y}_i) * (-1) = y_i - \hat{y}_i$$

- Initial model g_0 simply predicts the mean of y
- For iteration $m = 1..M$:
    - For all samples i=1..n, compute pseudo-residuals $r_i = y_i - \hat{y}_i$
    - Fit a new regression tree model $g_m(\mathbf{x})$ to r_i
        - In $g_m(\mathbf{x})$, each leaf predicts the mean of all its values
    - Update ensemble predictions $\hat{y} = g_0(\mathbf{x}) + \sum_{m=1}^{M} \eta \cdot g_m(\mathbf{x})$
- Early stopping (optional): stop when performance on validation set does not improve for nr iterations

Gradient Boosting Regression in action

- Residuals quickly drop to (near) zero



GradientBoosting Algorithm (Classification)

- Base models are *regression* trees, predict probability of positive class p
  - For multi-class problems, train one tree per class
- Use (binary) log loss, with true class $y_i \in {0,1}$:
  $\mathcal{L}_{log} = - \sum_{i=1}^{N} \left[y_i log(p_i) + (1-y_i) log(1-p_i) \right]$
- The pseudo-residuals are simply the difference between true class and predicted p:
  $$\frac{\partial \mathcal{L}}{\partial \hat{y}} = \frac{\partial \mathcal{L}}{\partial log(p_i)} = y_i - p_i$$
- Initial model g_0 predicts $p = log(\frac{\#positives}{\#negatives})$
- For iteration $m=1..M$:
  - For all samples i=1..n, compute pseudo-residuals $r_i = y_i - p_i$
  - Fit a new regression tree model $g_m(\mathbf{x})$ to r_i
    - In $g_m(\mathbf{x})$, each leaf predicts $\frac{\sum_i r_i}{\sum_i p_i(1-p_i)}$
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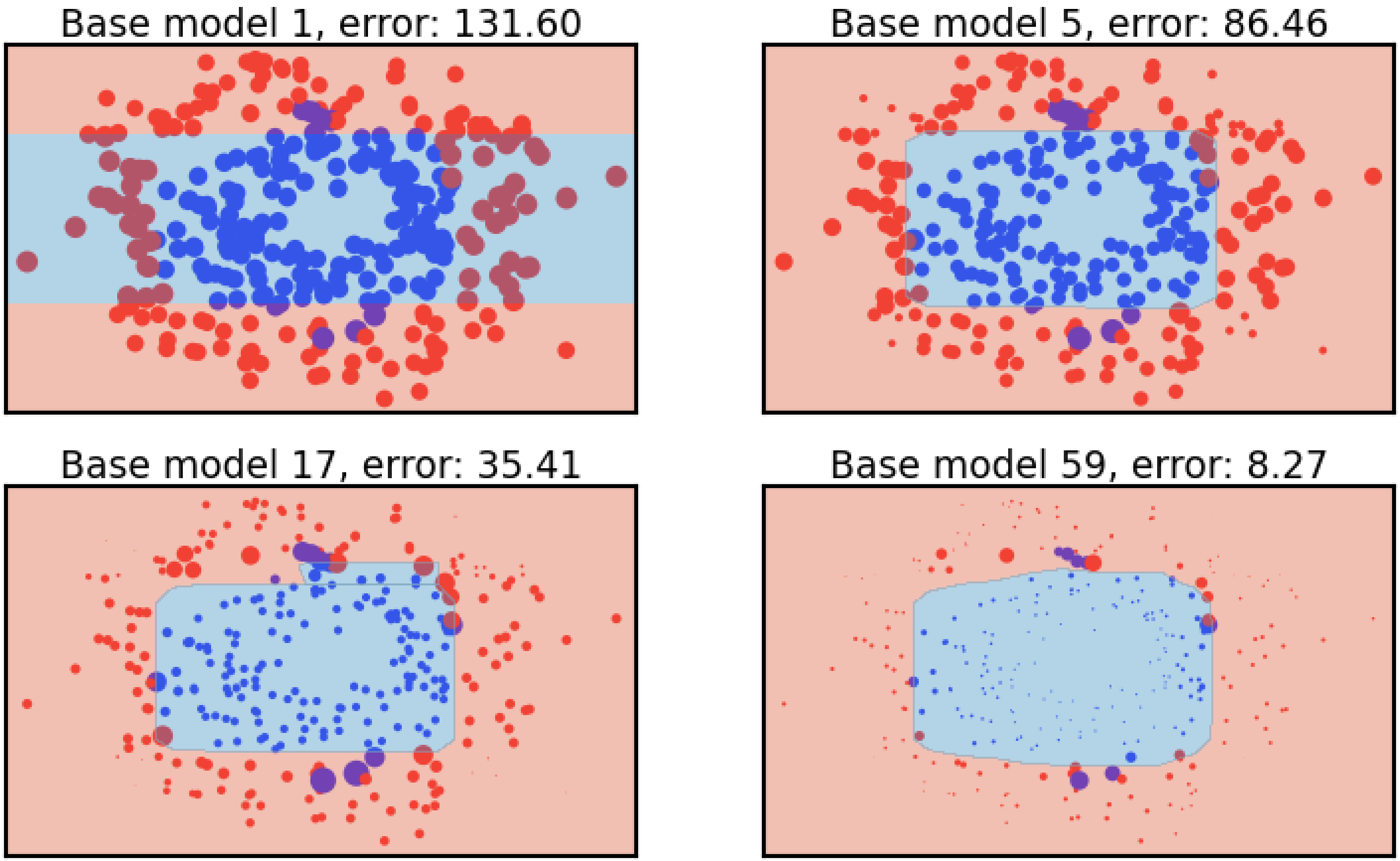
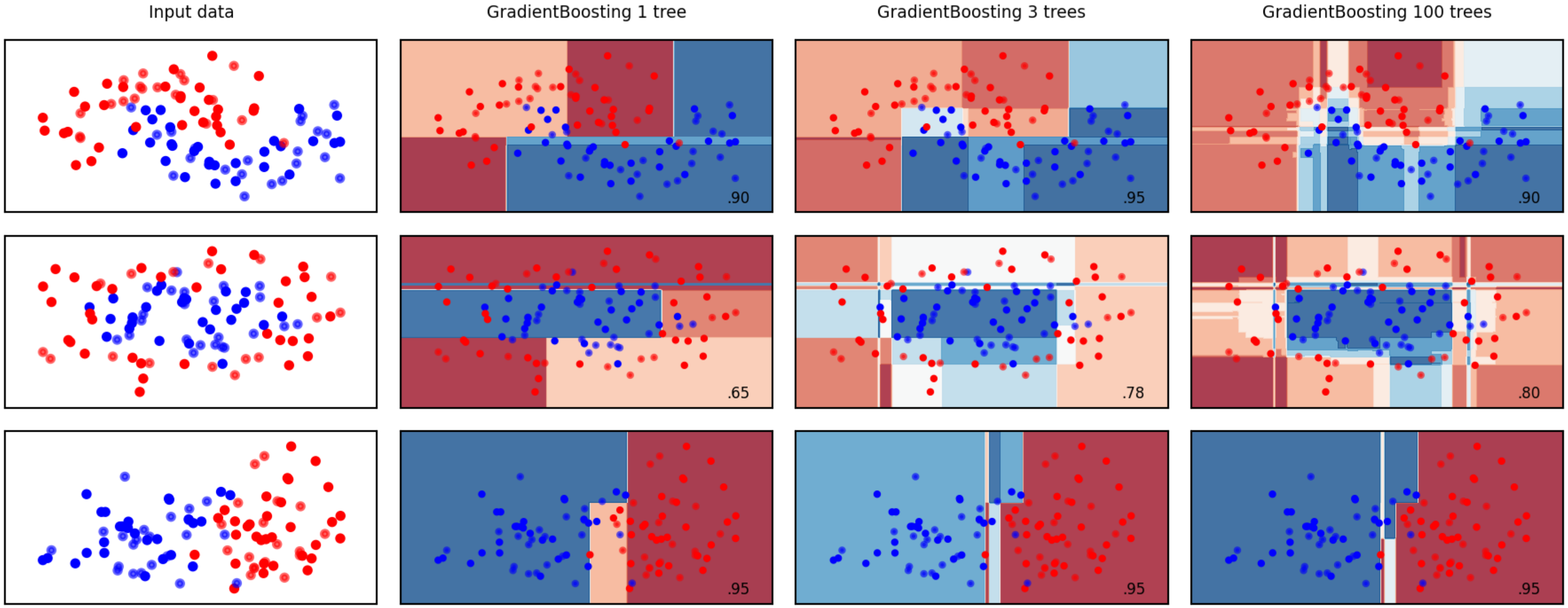
Gradient Boosting Classification in action

- Size of the samples represents the residual weights: most quickly drop to (near) zero



Examples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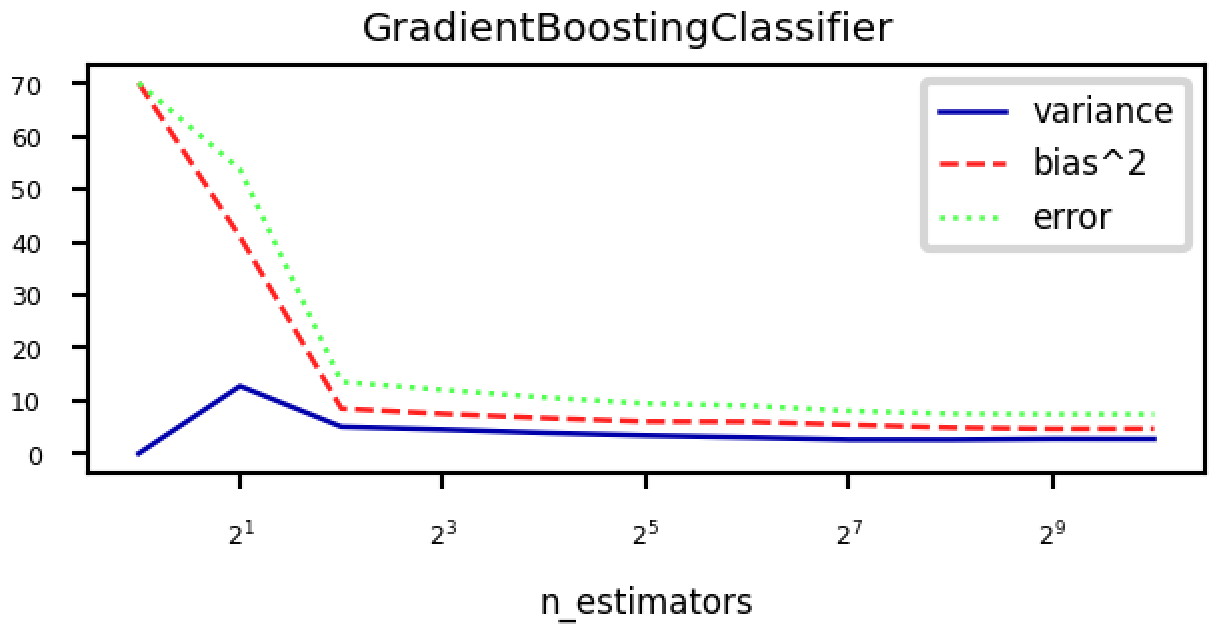
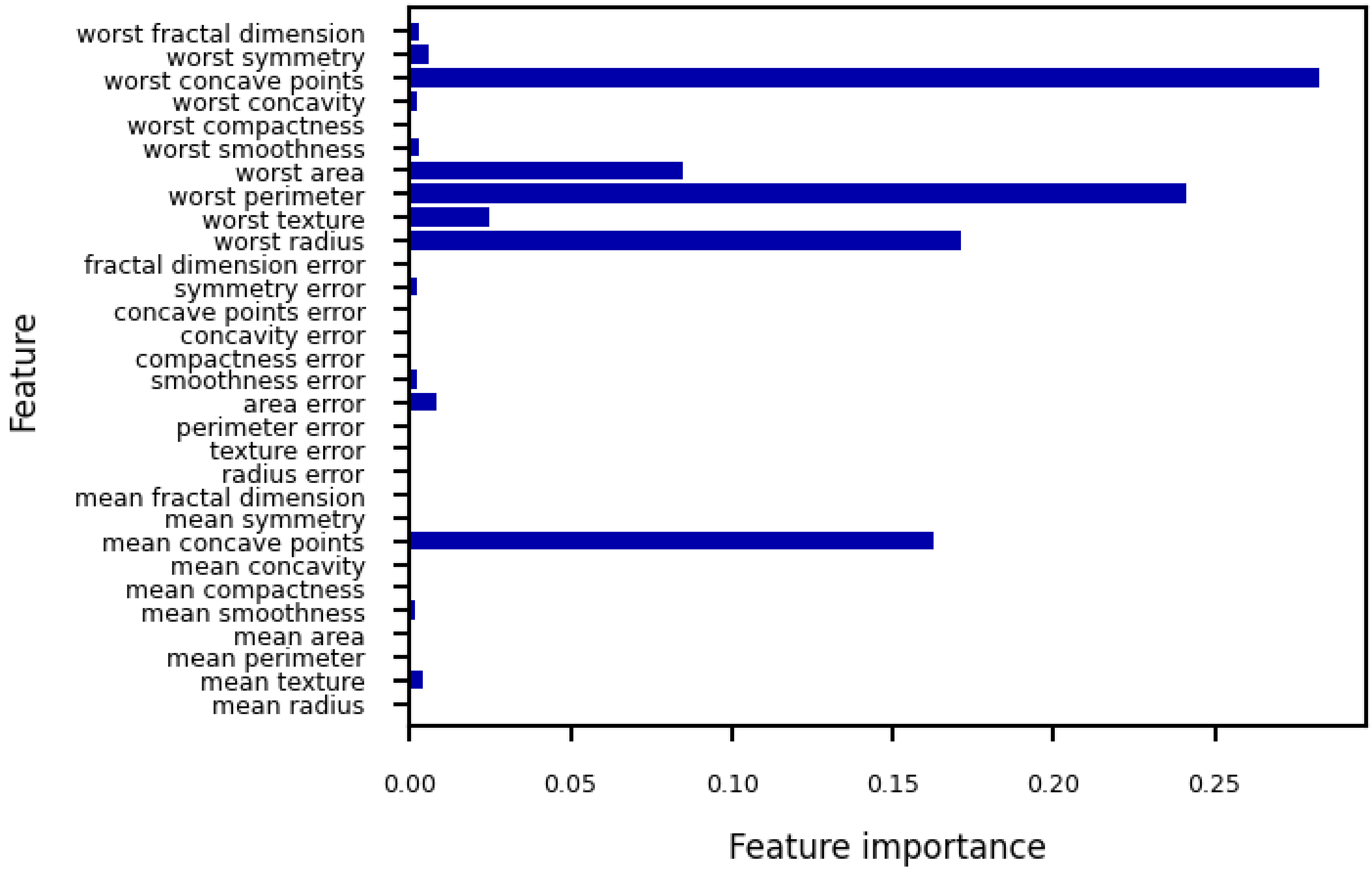
Bias-variance analysis

- Gradient Boosting is very effective at reducing bias error
- Boosting too much will eventually increase variance



Feature importance

- Gradient Boosting also provide feature importances, based on many trees
- Compared to RandomForests, the trees are smaller, hence more features have zero importance



Gradient Boosting: strengths and weaknesses

- Among the most powerful and widely used models
- Work well on heterogeneous features and different scales
- Typically better than random forests, but requires more tuning, longer training
- Does not work well on high-dimensional sparse data

Main hyperparameters:

- `n_estimators`: Higher is better, but will start to overfit
- `learning_rate`: Lower rates mean more trees are needed to get more complex models
    - Set `n_estimators` as high as possible, then tune `learning_rate`
    - Or, choose a `learning_rate` and use early stopping to avoid overfitting
- `max_depth`: typically kept low (<5), reduce when overfitting
- `max_features`: can also be tuned, similar to random forests
- `n_iter_no_change`: early stopping: algorithm stops if improvement is less than a certain tolerance `tol` for more than `n_iter_no_change` iterations.

Extreme Gradient Boosting (XGBoost)

- Faster version of gradient boosting: allows more iterations on larger datasets
- Normal regression trees: split to minimize squared loss of leaf predictions
    - XGBoost trees only fit residuals: split so that residuals in leaf are more *similar*
- Don't evaluate every split point, only q *quantiles* per feature (binning)
    - q is hyperparameter (`sketch_eps`, default 0.03)
- For large datasets, XGBoost uses *approximate quantiles*
    - Can be parallelized (multicore) by chunking the data and combining histograms of data
    - For classification, the quantiles are weighted by $p(1-p)$
- Gradient descent sped up by using the second derivative of the loss function
- Strong regularization by pre-pruning the trees
- Column and row are randomly subsampled when computing splits
- Support for out-of-core computation (data compression in RAM, sharding,...)

XGBoost in practice

- Not part of scikit-learn, but `HistGradientBoostingClassifier` is similar
    - binning, multicore,...
- The `xgboost` python package is sklearn-compatible
    - Install separately, `conda install -c conda-forge xgboost`
    - Allows learning curve plotting and warm-starting
- Further reading:
    - XGBoost Documentation
    - Paper
    - Video

LightGBM

Another fast boosting technique

- Uses *gradient-based sampling*
    - use all instances with large gradients/residuals (e.g. 10% largest)
    - randomly sample instances with small gradients, ignore the rest
    - intuition: samples with small gradients are already well-trained.
    - requires adapted information gain criterion
- Does smarter encoding of categorical features

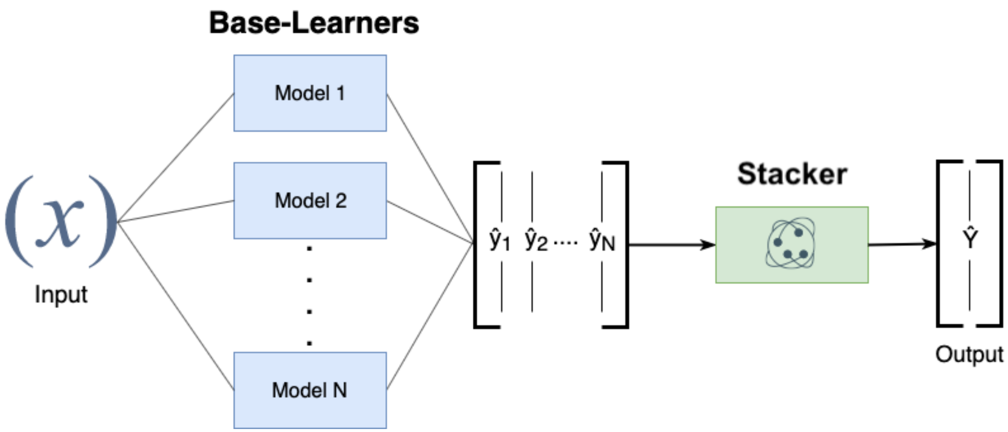
CatBoost

Another fast boosting technique

- Optimized for categorical variables
    - Uses bagged and smoothed version of target encoding
- Uses symmetric trees: same split for all nodes on a given level aka
    - Can be much faster
- Allows monotonicity constraints for numeric features
    - Model must be be a non-decreasing function of these features
- Lots of tooling (e.g. GPU training)

Stacking

- Choose M different base-models, generate predictions
- Stacker (meta-model) learns mapping between predictions and correct label
    - Can also be repeated: multi-level stacking
    - Popular stackers: linear models (fast) and gradient boosting (accurate)
- Cascade stacking: adds base-model predictions as extra features
- Models need to be sufficiently different, be experts at different parts of the data
- Can be *very* accurate, but also very slow to predict



Other ensembling techniques

- Hyper-ensembles: same basic model but with different hyperparameter settings
    - Can combine overfitted and underfitted models
- Deep ensembles: ensembles of deep learning models
- Bayes optimal classifier: ensemble of all possible models (largely theoretic)
- Bayesian model averaging: weighted average of probabilistic models, weighted by their posterior probabilities
- Cross-validation selection: does internal cross-validation to select best of M models
- Any combination of different ensembling techniques

Algorithm overview

Name	Representation	Loss function	Optimization	Regularization
Classification trees	Decision tree	Entropy / Gini index	Hunt's algorithm	Tree depth,...
Regression trees	Decision tree	Square loss	Hunt's algorithm	Tree depth,...
RandomForest	Ensemble of randomized trees	Entropy / Gini / Square	(Bagging)	Number/depth of trees,...
AdaBoost	Ensemble of stumps	Exponential loss	Greedy search	Number/depth of trees,...
GradientBoostingRegression	Ensemble of regression trees	Square loss	Gradient descent	Number/depth of trees,...
GradientBoostingClassification	Ensemble of regression trees	Log loss	Gradient descent	Number/depth of trees,...
XGBoost, LightGBM, CatBoost	Ensemble of XGBoost trees	Square/log loss	2nd order gradients	Number/depth of trees,...
Stacking	Ensemble of heterogeneous models	/	/	Number of models,...

Summary

- Ensembles of voting classifiers improve performance
    - Which models to choose? Consider bias-variance tradeoffs!
- Bagging / RandomForest is a variance-reduction technique
    - Build many high-variance (overfitting) models on random data samples
        - The more different the models, the better
    - Aggregation (soft voting) over many models reduces variance
        - Diminishing returns, over-smoothing may increase bias error
    - Parallellizes easily, doesn't require much tuning
- Boosting is a bias-reduction technique
    - Build low-variance models that correct each other's mistakes
        - By reweighting misclassified samples: AdaBoost
        - By predicting the residual error: Gradient Boosting
    - Additive models: predictions are sum of base-model predictions
        - Can drive the error to zero, but risk overfitting
    - Doesn't parallelize easily. Slower to train, much faster to predict.
        - XGBoost,LightGBM,... are fast and offer some parallellization
- Stacking: learn how to combine base-model predictions
    - Base-models still have to be sufficiently different